GUNESH GUPTA

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WORK EXPERIENCE

Indian Institute of Technology Roorkee (IIT Roorkee) Roorkee, Uttarakhand, India
Research Intern Feb 2026 - Present
Control and Autonomous Robotic Systems (CARS) Lab
2R-Manipulator link length optimization [arXiV(preprint)] demonstration

International Institute of Information Technology, Hyderabad (IIIT-H) Hyderabad, Telangana, India
Research Intern (at Robotics Research Center) Aug 2025 - Jan 2026

- Bipedal robotic locomotion using Reinforcement Learning within NVIDIA's Isaac Lab and Isaac Sim platforms.
- Sim to Sim transfer for biped locomotion from Isaac Lab to ROS2 Gazebo
  Biped Walking Demonstration Video (Isaac Lab and ROS2 Gazebo)

EDUCATION

Bennett University Greater Noida, UttarPradesh, India
Bachelor of Technology May 2025
Major in Computer Science and Engineering
Cumulative GPA: 9.23/10.00

PROJECTS

Self-Made PAROL6 (6 DOF robotic arm) May 2025

Built a six-degree-of-freedom robotic arm from scratch using custom electronics and ROS2 integration. Demonstrated the following features. Used Nema17s, DM542 drivers and Teensy 4.1 development board for controlling the robot. The features are itself links of the video demonstration:-

- GUI based control of each joint
- Multiple Joint Movement
- ROS2 Moveit2 integration
- Hand Gesture Control
- Sketching: Link 1, Link 2, Link 3

Autonomous Navigating Robot (SLAM with Nav2) March 2025

Made a differential drive 2 wheeled robot that with the help of odometry feedback using encoder motors and a A1M8 RPLIDAR applies SLAM toolkit to map any indoor environment. Tools used: ROS2, Gazebo, Rviz2, SLAM Toolkit. Demonstration:

- ROS2 Odometry Control
- Process for SLAM application
- SLAM with NAV2 (Following goal given in a MAP)
- NAV2 with waypoint following

6DOF UR5e Configuration and Movement with MOVEit2 Aug 2024

Configured the URDF and the MOVEit controller for Universal Robots' UR5e using ROS2(Humble), Gazebo and Rviz2. Simulation Link

Luminosity DRONE (E-YANTRA IIT Bombay) Feb 2024

My task was to find out the different types of aliens in a given exoplanet area and report back to the base station. Successfully completed the first 3 Tasks to qualify for Stage 2. In Stage 2 hardware implementation

of the previous tasks, I completed 3 out of the 4 Tasks given to us namely Task 3, 4, 5, and 6. Except for the last i.e. Task 6, all were successfully completed. Video Demonstration Links:

- PID Control
- Simulation of Full Task

3DOF Alexa Controlled Manipulator Arm August 2024

Following a course on Udemy, practically configured and made a 3 degree of freedom robotic manipulator that can be voice controlled through Amazon Alexa. Simulation Link

SOILEYE (Raspberry Pi, ESP32, 7 in 1 Soil Sensor) May 2023

Crop and fertilizer recommendation project. In this project our hardware model is capable of recommending to the farmer what crop to grow in his/her field and also the amount of fertilizer required in the field. Video Demonstration Link

RESEARCH PUBLICATION

Gunesh Gupta, Arvind Kumar Mishra, Sohom Chakrabarty "*Kinematic Reward, Measured Effort: Hardware-Validated RL Morphology Optimization of a Planar 2R Manipulator*" arXiV(preprint),
DOI: https://doi.org/10.48550/arXiv.2610.15756 October 2026

Manish Raj, Shakti Sharma, **Gunesh Gupta**, Shilpi Yadav "*Auto tuning PID using deep Q-learning*" IEEE 2025 International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC),
DOI: 10.1109/ASSIC64892.2025.11158657 May 2025

Debjani Ghosh, Jatin Swarnkar, **Gunesh Gupta**, Priyanka Rawat "*The Green Revolution from Space: A Survey of Satellite-Based Agricultural Monitoring Systems*" 2023 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS),
DOI: 10.1109/ANTS59832.2023.10468879 December 2023

SKILLS

C++, ROS2 (Humble), Python 3, Gazebo, Isaac Sim, Design in AutoCAD Fusion 360, Microcontrollers (Arduino, ESP32, Teensy 4.1), Single Board Computers (Raspberry Pi, NVIDIA Jetson Orin Nano)

CERTIFICATIONS

- Modern Robotics, Course 2: Robot Kinematics — Northwestern University through COURSERA. Credential ID: 5772SOVIAB9H
- Self Driving and ROS 2 - Learn by Doing! Map and Localization— By Antonio Brandi through UDEM. Credential ID: UC-2b0b0a53-8fc3-4610-a6d6-ec2d7d87690b
- Neural Networks and Deep Learning — By DeepLearning.AI (Dr. Andrew NG) through COURSERA. Credential ID: BFFA2WXDCMUE
- Machine Learning with Python — By IBM through COURSERA. Credential ID: ZJA477QZK8N6
- Beginning Custom Projects with Raspberry Pi with honors — By John Hopkins University through COURSERA Credential ID: RCS59HQ7THCT
- The Arduino Platform and C Programming— By University of California, Irvine through COURSERA Credential ID: 5TAW74R7EHKM